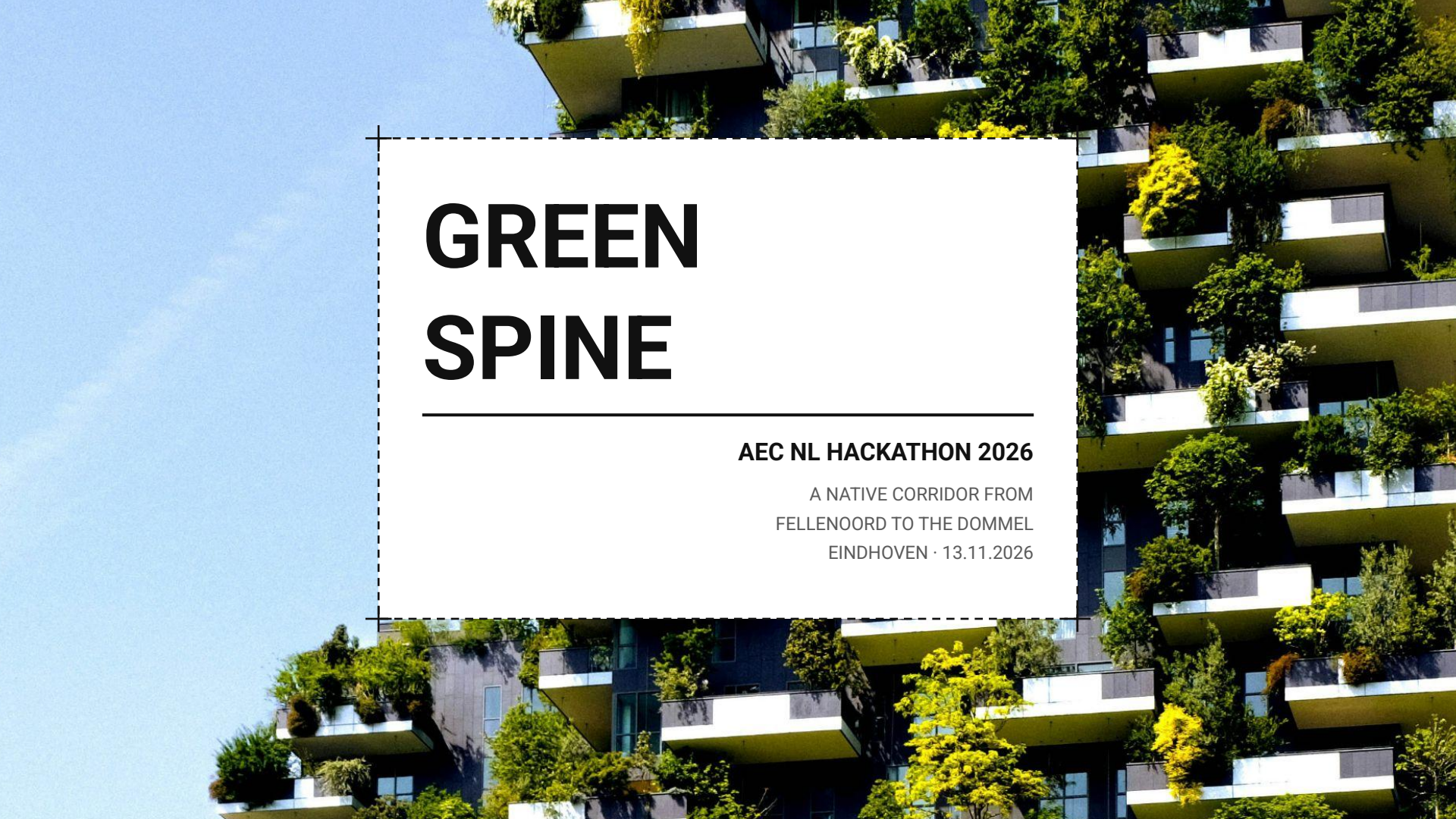


GREEN SPINE

AEC NL HACKATHON 2026

A NATIVE CORRIDOR FROM
FELLENOORD TO THE DOMMEL
EINDHOVEN · 13.11.2026

THE TEAM



RANIA CHIHAOUI

Architect and computational designer

Corridor design, placement model, visuals



QUIMARA VAN DEN HOEVEN

Structural engineering

Soil volumes, no-plant zones, KPI formulas



CORNE HAGEN

Structural engineering

Tree classes, buildability, Phasing, canopy and pergola



OLIWIA STANKIEWICZ

AI and data

Pipeline, scoring, species ranking, dashboard

THE STUDY CASE: FELLENOORD

249,783
people in Eindhoven, 1 January 2026

~1,000
homes on one acquired site, 85% affordable

6,000–7,000
homes planned on Fellenoord by 2040

30%
target share below the social rent threshold

A few hundred people live in the area today. By 2040 it becomes a dense city-centre district next to a station carrying 180,000 travellers a day. Eindhoven as a whole counts 40,676 residents under 18 and 40,771 over 65, the two groups most exposed to summer heat.

CBS via business.gov.nl (2026); citypopulation.de age groups (2025); Eindhoven News; BAM (2026); ProRail (2023).

Fellenoord is being rebuilt now. How do we distribute the vegetation?

01

NOW

Fellenoord is paved, road-dominated and about to be redeveloped. The window to decide where the green goes is open today

02

BELOW GROUND

Street trees die from too little rootable soil

03

UNMEASURED

Street greening projects state many goals; how to evaluate whether they are met?

Ontwikkelvisie Fellenoord; Toronto soil cell study (The Nature of Cities); Re-imagining Streets, University of Melbourne.

WHAT WE ARE BUILDING

“Based on the Fellenoord urban layout and parametric input, generate a dashboard with vegetation planting options optimizing the ecosystem, infrastructure impact, and canopy cover today, in 2040 and in 2060.”

GENERATES	Three planting strategies on the same corridor, from one scoring rule	**RESPECTS**	Soil, utilities and the carriageway. Trees only where the ground can carry them
MEASURES	Four KPIs at 2026, 2040 and 2060, against targets	**DELIVERS**	A planting schedule the city can act on, not only an image

Brundtland’s definition of Sustainability: Development that meets the needs of the present without compromising the ability of future generations to meet their own needs

WHO IT IS FOR

WHO	ROLE AT FELLENOORD	WHAT THEY WANT FROM THE SPINE
Gemeente Eindhoven	Owns the public space and the climate policy	Evidence that greening targets are met
Fellenoord 2040 BV	Development company for the city and province	A plan that works cluster by cluster to 2040
ProRail and NS	Station and bus interchange	Comfort for 180,000 daily travellers by 2040
City Planner	Need to defend where the planting budget goes, against targets, with evidence	A ranked planting schedule
Real Developers (AM, BAM)	Build the clusters, about 1,000 homes on one site	Clear green obligations per plot
Residents and pedestrians, now and 2040	Walk from the station to the Dommel	A continuous shaded route to take in summer
Animals	Insects, small animals and pets	A safe, walkable zone supporting their wellbeing

Ontwikkelvisie Fellenoord; Provincie Noord-Brabant; ProRail (2023); BAM press release (July 2026).

DATA COLLECTION

INPUT DATA

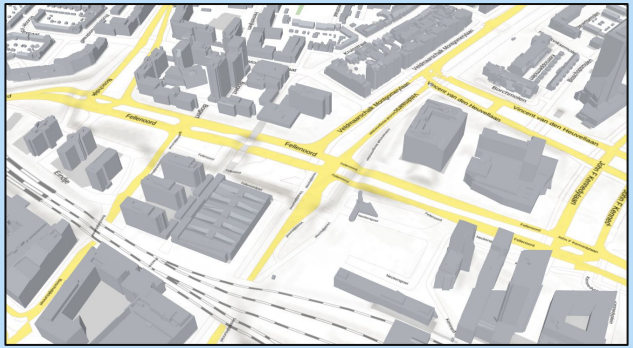
ALL INPUTS

11 layers feed the model

1. **Buildings with heights**
2. **Trees**
   - Roads
   - Green areas
   - The Dommel and wet ground
   - Corridor and walking line
3. **Soil boreholes**
4. **Plant database**
   - Native species list
   - Sun position
   - Policy targets

1 BUILDINGS WITH HEIGHTS

3D BAG: footprint and height of every building



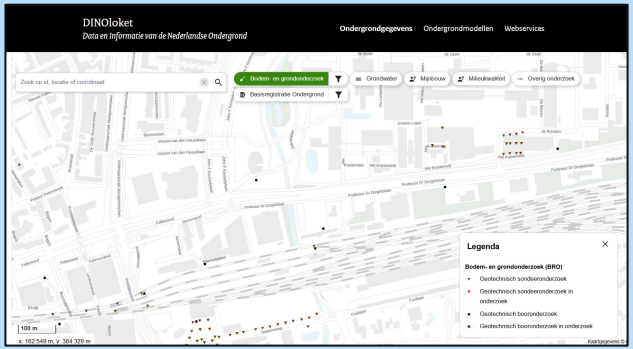
2 TREES

Eindhoven open data: every municipal tree as a point



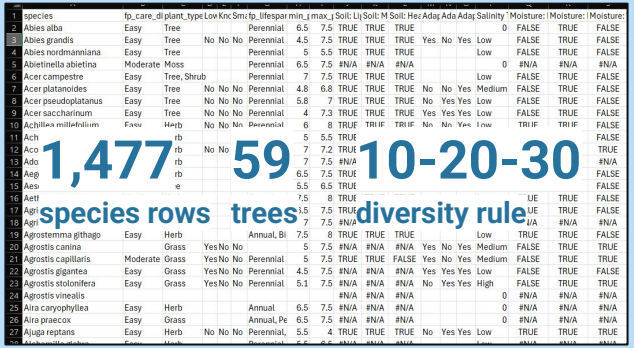
3 SOIL

Eindhoven atlas: soil types (boreholes next)



4 PLANT DATABASE

Team's merged species table



3dbag.nl; data.eindhoven.nl (Bomen); Eindhoven atlas, eindhoven.nazca4u.nl; team plant database.

RAW DATA AND HOW WE COLLECT IT

LAYER	SOURCE
Plant database	Merged species table from GBIF, POWO, PlantNet, PFAF and ForwardPlant
Buildings in 3D	3D BAG, four CityJSON tiles
Policy and KPIs per plot	Ontwikkelvisie and cluster passports 1A, 1B, 2, 9B, 9C
Existing trees	Eindhoven open data, Bomen, public domain
Public green	Eindhoven open data, Openbaar groen
Surfaces and roads	OpenStreetMap
Soil and heat	DINOloket and OpenStreetMap: No open layer at street resolution

data.eindhoven.nl; 3dbag.nl; OpenStreetMap contributors (ODbL); Klimaateffectatlas (download options limited).

HOW WE USED DATA

HOW WE USE THE DATA

1	**CLIP AND ALIGN**	Every layer cut to the corridor and converted to the Dutch national grid, EPSG:28992, so metres are metres
2	**GRID**	The corridor's public space divided into 2 m cells
3	**ATTRIBUTE**	Each cell inherits blocked, paved, soil, heat and distance to existing canopy
4	**ONE FILE**	All of it becomes a single scored grid, the only handover between the data work and the design model

Utilities, traffic and pedestrian flow are stated assumptions, not hidden guesses.

KPIs AND METRICS

KPI	HOW WE CALCULATE IT	TARGET
Canopy cover	Merged crown area divided by corridor area	30% (3-30-300 rule)
Shaded route	Walking centreline in shade, divided by total length	60% (our proposal)
Longest gap	Longest uncovered stretch of the centreline, in metres	25 m or less (our proposal)
Trees in natural soil	Trees that do not need engineered soil, divided by all trees	60% (our proposal)
Species diversity	Share of the largest species in the cluster	10-20-30 rule

Also reported: tree count, species count and engineered soil volume in m^3 as a cost proxy.

3-30-300 rule (Konijnendijk); 10-20-30 rule (Santamour). Shade, gap and soil targets are our own proposals.

ALGORITHMIC WORKFLOW

INPUTS

3D mass, surfaces, greenery, soil, climate, plants

→

SCORE

Each cell: heat + soil + gap to green

→

PLACE

Best cell first, largest tree the soil allows, spacing by crown

→

SPECIES

Filter by soil and size, enforce 10-20-30

→

MEASURE

Four KPIs at 2026, 2040, 2060

LOOP: rerun with three weight sets to generate options A, B and C, then rank by targets met

A transparent scoring rule, not a black box. Every tree can be traced back to the cell score that put it there.

SCOPE, ASSUMPTIONS, LIMITATIONS

- Cost estimation
- Plant database only covers north brabant
- No water retention

DEMO